



Master Presenter's Manual

for

Developing Microsoft Azure Solutions

Curriculum Code: OV-1510

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Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	April 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Work with Microsoft Azure cloud platform and its architecture
- Create and deploy cloud services
- Create and use virtual machines
- Explore Service Fabric and Service Bus
- Design and Implement a Web API service
- Secure a Web API service
- Host and Consume API Web Services
- Access and Manipulate Data
- Query Data with Entity Framework
- Understand Characteristics of Azure Storage
- Understand Azure Storage Services and Use Storage Accounts
- Create, host and consume WCF services
- Work with Azure SQL Database
- Understand Azure Containers

2. Information on Module Allocation

Module	Online (No. of Hours)	Skill Code
Azure	40	1341

Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

3. Week-wise Session Schedule

- A session is of 2 hours duration.

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 AZURE-TL1	Session 2 AZURE-TL2	Session 3 AZURE-TL3	Session 4 AZURE-TL4
2	Session 5 AZURE-TL5	Session 6 AZURE-TL6	Session 7 AZURE-TL7	Session 8 AZURE-TL8
3	Session 9 AZURE-TL9	Session 10 AZURE-TL10	Session 11 AZURE-TL11	Session 12 AZURE-TL12
4	Session 13 AZURE-TL13	Session 14 AZURE-TL14	Session 15 AZURE-TL15	Session 16 AZURE-TL16
5	Session 17 AZURE-TL17	Session 18 AZURE-TL18	Session 19 AZURE-TL19	Session 20 AZURE-TL20

Azure: Developing Microsoft Azure Solutions
TL: Online Session

4. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	Azure – TL1	All the topics as listed below from Session 1 and Session 2 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session. <u>Session 1: Introduction to Azure Cloud Computing</u> • Microsoft Azure Cloud Services • Azure Availability and SLAs • Azure cloud computing • Features and Functionalities of Azure Cloud Computing <u>Session 2: Understanding Microsoft Azure Cloud Services</u> • Virtual Machines (VM) Overview • Creating, Configuring, and managing VMs in Azure • Deploying a VM in Azure • Data and Data Access Technologies • Advanced Azure Cloud Services	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> SG - Session 1 & 2 XP - Session 1 & 2 TG - Session 1 & 2
2	Azure – TL2	All the topics as listed below from Session 3 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session. <u>Session 3: Azure Management Tools</u> • Azure CLI • Azure PowerShell • Azure Cloud Shell • Azure Resource Manager	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> SG - Session 3 XP - Session 3 TG - Session 3

Session No.	Session Title	Session Details	Deliverables' Mapping
3	Azure – TL3	All the topics as listed below from Session 4 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session. <u>Session 4: Azure Monitoring</u> • Azure Monitoring • Azure Monitor Application Insights • Monitoring solutions in Azure Monitor • Logs in Azure Monitor • Finding and Diagnosing run-time exceptions with Azure Application Insights	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> SG - Session 4 XP - Session 4 TG - Session 4
4	Azure – TL4	The Try It Yourself questions of Session 1 to Session 4 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session.	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> Session 1-4
5	Azure – TL5	All the topics as listed below from Session 5 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session. <u>Session 5: Using Entity Framework in Azure</u> • Entity Framework Core 8.0 • Design Approach in Entity Framework • Database Operation in Entity Framework • Complex Query in LINQ • Accessing data using EF	SG - Session 5 XP - Session 5 TG - Session 5
6	Azure – TL6	All the topics as listed below from Session 6 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session. <u>Session 6: Working With Data In Azure Applications</u> • Injecting data from various	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> SG - Session 6 XP - Session 6 TG - Session 6

Session No.	Session Title	Session Details	Deliverables' Mapping
		sources to Azure • Application of Azure SQL Database, Azure Cosmos DB, Azure Data Lake Storage, Azure Synapse Analytics, and Azure Databricks • Implementation of data caching services in Azure • Distributed transactions in SQL Azure • Role of Content Delivery Networks (CDNs) • Master data processing using Azure services such as Azure HDInsight, Azure Databricks, and Azure Stream Analytics for batch and real-time data processing	
7	Azure – TL7	All the topics as listed below from Session 7 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session. <u>Session 7: Design and Implementation of Web API</u> • Implementing Web API • Routing in Web API • Implementing Database Operation Using Web API • Implementing Asynchronous and Synchronous Actions • Creating OData Services	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> SG - Session 7 XP - Session 7 TG - Session 7
8	Azure – TL8	The Try It Yourself questions of Session 5 to Session 7 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session.	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> Session 5-7
9	Azure – TL9	All the topics as listed below from Session 8 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session. <u>Session 8: APIs and RESTful APIs for Enterprise Applications</u> • Implementing Authentication	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> SG - Session 8 XP - Session 8 TG - Session 8

Session No.	Session Title	Session Details	Deliverables' Mapping
		Mechanisms - Setting Up Role-Based Access Control (RBAC) - Versioning Strategies - HATEOAS (Hypermedia as the Engine of Application State)	
10	Azure – TL10	All the topics as listed below from Session 9 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session. <u>Session 9: Azure Security Features</u> - Securing networks - KeyLogs ,Key vault - Malware protection, Access Management - Configuring MFA for Azure AD users and administrators - Azure Firewall and NSGs usage to control network traffic and protect resources. - Configuring and monitor DDoS protection for your resources	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> SG - Session 9 XP - Session 9 TG - Session 9
11	Azure – TL11	All the topics as listed below from Session 10 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session. <u>Session 10: Hosting and Consuming Web API</u> - Hosting Web API in Windows Azure Web App - Hosting Web API on Containers Services - Consuming Web API - Secure hosted API Like AAD	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> SG - Session 10 XP - Session 10 TG - Session 10
12	Azure – TL12	The Try It Yourself questions of Session 8 to Session 10 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session.	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> Session 8-10
13	Azure – TL13	All the topics as listed below from Session 11 of <i>Build Enterprise and Cloud Applications with Microsoft</i>	<u>Build Enterprise and Cloud Applications with</u>

Session No.	Session Title	Session Details	Deliverables' Mapping
		Azure book should be covered in this session. <u>Session 11: WCF Services</u> • ABC of WCF Services • Creating and Deploying WCF Services • Configuring WCF Services • Creating and Configuring Bindings for WCF Services • Integrating with the Azure Service Bus Relay • Generating Proxies Using SvcUtil • WCF Client • Generating Proxies by Creating a Service Reference • WCF Security • Containerize WCF services and host them in Azure Kubernetes • Implement scaling based on metrics and rules • Set up high-availability solutions for WCF services to ensure business continuity	<u>Microsoft Azure</u> SG - Session 11 XP - Session 11 TG - Session 11
14	Azure – TL14	All the topics as listed below from Session 12 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session. <u>Session 12: Implement Azure Serverless Web Services And API Management</u> • What is API gateway pattern • Secure Web Services using Certificates, Azure Active Directory, and OAuth • Provision an API function • Creating a new revision • Configuring API policies • Understanding scopes and where to apply them • Serverless API workshop • API Interface using the Azure Portal and Swagger • Managing Running Services	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> SG - Session 12 XP - Session 12 TG - Session 12

Session No.	Session Title	Session Details	Deliverables' Mapping
		(using logging, disaster recovery, and multiple regions) - Mastering API design principles, including RESTful design, versioning, and content negotiation - Implementing event-driven architectures using Azure Functions	
15	Azure – TL15	The Try It Yourself questions of Session 11 and Session 12 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session.	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> Session 11 & 12
16	Azure – TL16	All the topics as listed below from Session 13 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session. <u>Session 13: CI/CD</u> - CI/CD - CI/CD with Azure - Integrating with Azure DevOps and other CI/CD tools - Setting up artifact repositories - IaC concepts using Azure Resource Manager templates, Terraform, or other declarative scripting languages	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> SG - Session 13 XP - Session 13 TG - Session 13
17	Azure – TL17	All the topics as listed below from Session 14 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session. <u>Session 14: Implementing Traffic Management and Monitoring Strategies for Web Services</u> - Introduction to Azure Load-balancer - Working with Azure Application Gateway and Traffic Manager - Introduction to Azure Application Insights and Log Analytics - Configuring Application Insights	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> SG - Session 14 XP - Session 14 TG - Session 14

Session No.	Session Title	Session Details	Deliverables' Mapping
		and OMS (to view the logs for the Web applications) • Introduction to Azure Event Hubs and Stream Analytics • Configuring routing and caching options for web services using Front Door • Setting up monitoring and telemetry solutions using Azure Monitor and Application Insights	
18	Azure – TL18	The Try It Yourself questions of Session 13 and Session 14 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session.	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> Session 13 & 14
19	Azure – TL19	All the topics as listed below from Session 15 of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session. <u>Session 15: Deploying Web Applications and Azure App Service Apps</u> • Setting Up Deployment Center ◦ Github, Docker, Bitbucket, OneDrive • Understand deployment • Publish files deployment • Designing Continuous Deployment Pipeline • Working with Deployment Slots for Web apps	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> SG - Session 15 XP - Session 15 TG - Session 15
20	Azure – TL20	The Try It Yourself questions of Session 15 and Appendix of <i>Build Enterprise and Cloud Applications with Microsoft Azure</i> book should be covered in this session.	<u>Build Enterprise and Cloud Applications with Microsoft Azure</u> Session 15

5. Deliverables

The deliverables to be provided to the students and the faculty have been tabulated below:

➤ **Deliverables for Student:**

Module	Deliverable
Developing Microsoft Azure Solutions	Kit Code: OV-1510-KIT01-01-A Kit Name: Kit of 1 title for OV-1510 Enterprise Appln Dev in DOTNET Book Code: AINNENTAZ10324E000 Book Name: Build Enterprise Cloud Applications with MS Azure

Deliverables for Faculty:

Module	Deliverable
Developing Microsoft Azure Solutions	Kit Code: OV-1510-KIT01-01-A Kit Name: Kit of 1 title for OV-1510 Enterprise Appln Dev in DOTNET Book Code: AINNENTAZ10324E000 Book Name: Build Enterprise Cloud Applications with MS Azure
Course	Classroom Presentations Trainer Guide Master PM

6. Library References

➤ Web Applications on Azure - Developing for Global Scale by Reagan, Rob
➤ Practical Azure Application Development A Step-by-Step Approach to Build Feature-Rich Cloud-Ready Solutions by Vijayakumar, Thurupathan

7. Evaluation Strategy

The Evaluation Strategy for this course is as follows:

- Final Examination

7.1 Award of Grades

Overall Weighted Marks $\geq 40\%$ but $< 60\%$ qualifies for **PASS**
Overall Weighted Marks $\geq 60\%$ but $< 75\%$ qualifies for **CREDIT** and
Overall Weighted Marks $\geq 75\%$ qualifies for **DISTINCTION**

Note: To attain a PASS/CREDIT/DISTINCTION grade, a student should achieve at least 40% in Final Examination; otherwise he/she will be declared as 'Referred'.

7.2 Final Examination

Below is the list of modules to be covered in Final Examination along with their weightages:

Final Examination (E1)

Marks: 50
Duration: 60 minutes
Type: Objective

Sr. No.	Name of the Subject	Distribution of Marks
1.	Developing Microsoft Azure Solutions	50

Final Examination (R1)

Marks: 50
Duration: 120 minutes
Type: Practical

Sr. No.	Name of the Subject	Distribution of Marks
1.	Developing Microsoft Azure Solutions	50

Evaluation Details

Practical – 50%
OES – 50%

8. Classroom and Lab Pre-requisites

The requirements for conducting the module of this course have been described as follows:

8.1 Software Requirements

Microsoft Windows 10 or higher
Microsoft Visual Studio 2022 Professional Edition

8.2 Recommended Hardware

Intel Core i5 Processor or higher
8 GB RAM or above
Color SVGA
500 GB Hard Disk space
Mouse
Keyboard

8.3 Other Requirements (For Classroom)

Computer for giving demonstrations wherever applicable
LCD projector

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